

Ayesha Ahmed

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Applied economist with experience designing randomized experiments, conducting causal inference, and analyzing large observational datasets. Research spans financial decision-making, credit markets, consumer demand, and policy evaluation using Python, R, and Stata. Interested in economist, experimentation, and data science roles focused on causal, policy, and product analysis.

EDUCATION

University of California, Davis Davis, CA
Ph.D. in Agricultural and Resource Economics (Applied Economics) Sep 2021–Jun 2027 (Expected)
Fields: Development, Behavioral & Experimental Economics, Credit Markets & Risk Management
Dissertation: Behavioral Determinants of Repayment and Credit Rationing in Microfinance

University of Dhaka Dhaka, Bangladesh
M.S.S. in Economics (CGPA: 3.88/4.00, top 2%) Jan 2019–Dec 2019
B.S.S. in Economics (CGPA: 3.96/4.00, top 1%) Jan 2015–Dec 2018

EXPERIENCE

University of California, Davis Davis, CA
Graduate Student Researcher Aug 2022–Present

- Contribute to the design of a randomized controlled trial with 6,683 agricultural–microfinance clients evaluating indexed financial tools by translating research questions into context-specific field designs and survey instruments; lead field implementation and enumerator training, coordinate with research partners, and use interim analyses to refine instruments and protocols.
- Designed and led a lab-in-the-field experiment with agricultural–microfinance clients in Bangladesh; developed the measurement strategy, led field implementation, and estimated treatment effects and conducted full analysis.
- Developed a survey based framework for credit constraints measurement; designed instruments and analyzed borrower level measures to support experimental design and evaluation.

Invited Lectures 2024–2026

- Designing and implementing lab-in-the-field experiments, and field protocols and findings from Bangladesh (PhD methods sequence); research communication (undergrad thesis track).

Teaching Assistant Sep 2021–Jun 2023

- Led weekly discussion sections and office hours for econometrics, microeconomics, and corporate finance courses with 30–180 students; prepared instructional materials and communicated quantitative concepts to students with varying technical backgrounds.

BRAC Institute of Governance and Development (BIGD) Dhaka, Bangladesh
Research Associate Aug 2020–Aug 2021

- Led research protocol implementation for an impact evaluation of a skills program for disadvantaged youth; designed questionnaires, trained enumerators, supervised field surveys, and cleaned and analyzed survey data.
- Analyzed household migration data and contributed to research reports, including a chapter on youth migration for BRAC's Urban Development Program.

BRAC James P. Grant School of Public Health Dhaka, Bangladesh
Research Assistant Feb 2020–Jul 2020

- Conducted quantitative analysis and co-authored the policy report on the health system response to COVID-19.
- Designed survey instruments, conducted the primary analysis, and contributed to writing a chapter on the social return on investment of BRAC's Health, Nutrition, and Population Program.

Bangladesh Institute of Development Studies (BIDS) Dhaka, Bangladesh
Research Assistant Aug 2019–May 2020

- Constructed price-volatility index from agricultural commodity price datasets and applied empirical and GIS methods to analyze crop diversification trends in Bangladesh.

University of Dhaka, Department of Economics Dhaka, Bangladesh
Research Assistant Sep 2019–Dec 2019

- Prepared analysis for a gender-responsive budget report for the Ministry of Health and Family Welfare.

SKILLS & METHODS

Programming: Python (pandas, scikit-learn, statsmodels, data pipelines, string matching), Stata, R, Julia, SQL
Econometrics & Causal Inference: Difference-in-differences, instrumental variables, event studies, synthetic control, panel data econometrics, experimental design
Machine Learning & NLP: Regularization, neural networks, tree-based and ensemble methods; text preprocessing, topic modeling, sentiment analysis
Tools & Platforms: VS Code, L^AT_EX, Git, SurveyCTO, ArcGIS

Additional Training: *Supervised Machine Learning: Regression and Classification*, Stanford & DeepLearning.AI; *Advanced Learning Algorithms*, Stanford & DeepLearning.AI; *Text Analysis and Natural Language Processing for Data Science*, UC Davis DataLab

Languages: Bangla (native), English (fluent), Spanish, Hindi (conversational)

WORKING PAPERS

Ahmed, A. 2026. “Audience Effect in Microfinance: Does Social Observability of Default Increase Repayment?” (Job Market Paper)

Does making default observable to peers increase repayment in uncollateralized loans? A framed lab-in-the-field experiment with 919 agricultural borrowers across 100 borrowing groups in Bangladesh varies whether default on a hypothetical microfinance loan is observable to peers. The within-subject design identifies the audience effect and separates intrinsic from reputational motivation to repay. An audience raises repayment by 3.3–5.3 percent, with no variation detectable across income states; a field reduction in default of the same size would raise lender profit by 34–55 percent of the zero-default margin.

Mahzab, M., **Ahmed, A.**, Brooks, N., Mahajan, A., Luby, S., Biswas, D., Ahmed, R., and Ahmed, S. 2026. “When Corruption Is the Norm: Can Signaling Environmental Compliance Reduce Bribery?”

Using randomized assignment to a broader environmental technology intervention as an instrument for banner display, we estimate the causal effect of compliance signaling on illicit payments per 1,000 bricks in Bangladesh’s brick-kiln sector. Contrary to the concern that advertising compliance invites rent extraction, each additional week of banner display lowers bribe payments by 18.6 percent and compliance costs by 5.4 percent.

RESEARCH IN PROGRESS

“Minimum Wages and Consumer Demand: Evidence from U.S. Fast-Food Chains” with Modhurima Dey Amin, Syed Badruddoza

Examines how minimum wage regulation affects consumer spending and store revenue in the U.S. fast-food industry. A triple differences (DDD) design exploits variation across states, over time, and between chains covered and not covered by California’s fast-food minimum wage law (AB 1228) across a high-frequency panel of 680,000+ store-level sales and customer-traffic observations. Data processing and analysis conducted in Python.

“Repayment Motivations and Financial Behaviors in Microfinance: Do Experimental Measures Predict Field Decisions?”

Estimates individual-level measures of intrinsic and reputational repayment motivation from repeated choices in the framed experiment, predicting latent borrower behavior and audience responsiveness with random-effects and two-limit Tobit specifications while accounting for measurement error in the recovered behavioral responses. The resulting measures are then tested against subsequent field outcomes, including insurance uptake and non-price credit rationing—an out-of-sample check on whether a behavioral measure built under experimental conditions predicts real decisions.

“Indexed Financial Tools and Risk Rationing in Microfinance” with Michael Carter

Does mitigating production risk address credit rationing for farmers? A large-scale randomized experiment with 6,683 farmers compares standalone agricultural index insurance against insurance bundled with savings and a contingent line of credit, measuring effects on different forms of non-price credit rationing. The design also tests whether risk rationing is more prevalent among borrowers more responsive to reputational incentives—that is, whether social collateral itself discourages borrowing under risk.

“Measuring Non-Price Credit Rationing in Microfinance”

Measures the prevalence and composition of non-price credit rationing among microfinance borrowers. We adapt the Direct Elicitation Method to the microfinance setting to separate borrowers who cannot obtain credit (quantity rationed), those deterred by its price (price rationed), and those who decline credit they could get in order to avoid risk (risk rationed). Provides the measurement framework for studying how exposure to risk shapes credit demand and access.

GRANTS, FELLOWSHIPS & AWARDS

Jastro & Shields Graduate Research Award, UC Davis (\$3,000)	2026
Subgrant , Resilience+ Innovation Facility, UC Davis (\$30,740)	2024
UC Davis Internal Fellowships & Awards (\$10,750 total)	2021–2026
Shamsun Nahar Mahmud Foundation Merit Award, University of Dhaka	2019
Governor’s Scholarship, University of Dhaka	2015–2017

CONFERENCE PRESENTATIONS

Northeastern Universities Development Consortium; AAEA Annual Meeting, MO; Symposium in Honor of Michael Carter; Pacific Conference for Development Economics	2026
AAEA & WAEA Joint Annual Meeting, CO (Poster); Bay Area Behavioral Economics and Experimental Workshop	2025
SANEM–World Bank North America Discussion Forum, Washington, DC	2023
15th South Asian Economics Students’ Meet, World Bank (Best Paper); SANEM Annual Economists’ Conference	2019
14th South Asian Economics Students’ Meet, World Bank; 1st Bangladesh Economics Summit, University of Dhaka (Best Paper); Speakonomics at EconFest, North South University (Best Paper)	2018